Beyond the Fundamental Wilson Line

New Operators and RG Flows of Wilson Lines in General Representations

Daniele Artico with C. Meneghelli, M. Savi and R. Treilis

Università di Parma & INFN, Gruppo Collegato di Parma

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D. Artico, C. Meneghelli, M. Savi, R. Treilis,

“New exotic operators in the spectrum of Wilson lines in general representations”,

arXiv:2606.07738 [hep-th].

Longer companion papers in preparation: algebraic aspects of the line; perturbative bootstrap and deformations.

D. Artico, J. Barrat, G. Peveri,
"Perturbative bootstrap of the Wilson-line defect CFT: multipoint correlators",
JHEP 02 (2025) 190, arXiv:2410.08271.

D. Artico, J. Barrat, Y. Xu,
"Perturbative bootstrap of the Wilson-line defect CFT: bulk-defect-defect correlators",
JHEP 03 (2025) 191, arXiv:2410.08273.

Also: D. Artico, PhD thesis, arXiv:2602.09226.

Outline

1 **Wilson lines as conformal defects**

Universality, topological and conformal defects, the Wilson line in $\mathcal{N} = 4$ SYM

2 **New operators on the line**

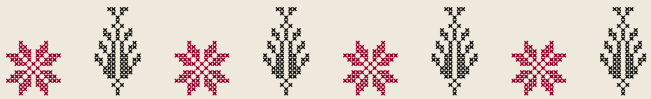
An algebra of insertions and its Hilbert series

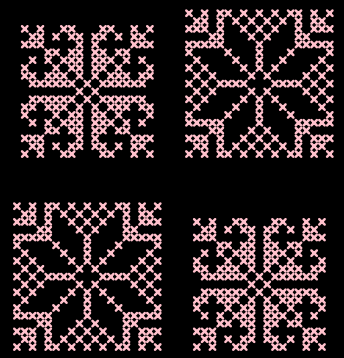
3 **Deforming the line**

Beta functions from defect OPE data

4 **Beyond the fundamental representation**

RG flows from the ordinary line, and new directions





WILSON LINES AS CONFORMAL DEFECTS

At a critical point, very different systems flow to the same scale-invariant theory

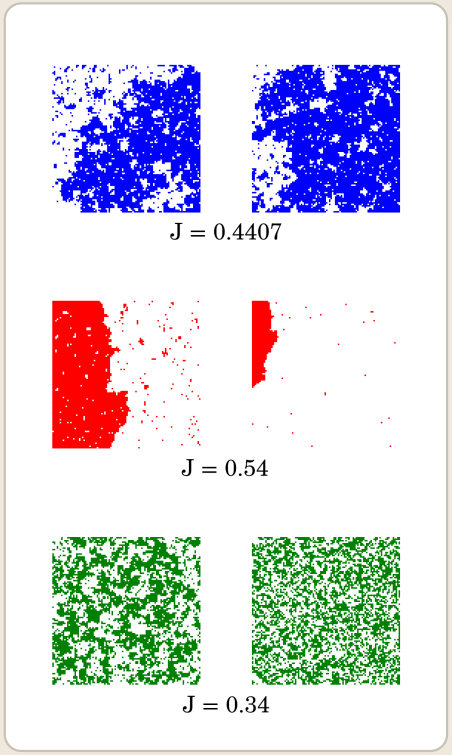
[Wilson, Kogut, '74][Cardy, '96]

Zooming out we flow along the renormalisation group. Near a fixed point K^*,

$$K'_a - K^*_a = \sum_b T_{ab} \left(K_b - K^*_b\right),$$

and the eigenvalues of T sort couplings into relevant, marginal and irrelevant.

Fixed points are scale invariant, and in practice conformal: CFTs label universality classes. Liquid–gas and magnetic transitions share the same exponents.



2d Ising: at, above, below J_c

Seen from far away, every cow is spherical: the endpoints of RG flows are CFTs.

A CFT is specified by its data, and consistency of the operator expansion constrains them

[Polyakov, '74][Rattazzi, Rychkov, Tonni, Vichi, '08][Poland, Rychkov, Vichi, '19]

Local operators have dimensions Δ_i, and their product is a convergent expansion:

$$\mathcal{O}_i(x)\,\mathcal{O}_j(0) = \sum_k \frac{\lambda_{ijk}}{|x|^{\Delta_i+\Delta_j-\Delta_k}} \left(\mathcal{O}_k(0) + \text{descendants}\right).$$

The data $\{\Delta_i, \lambda_{ijk}\}$ fix all correlators. Different ways of expanding the same four-point function must agree: crossing.

- ✕ With unitarity, crossing gives the 3d Ising exponents to many digits.
- ✕ Correlators are also special functions: polylogarithms, and beyond.

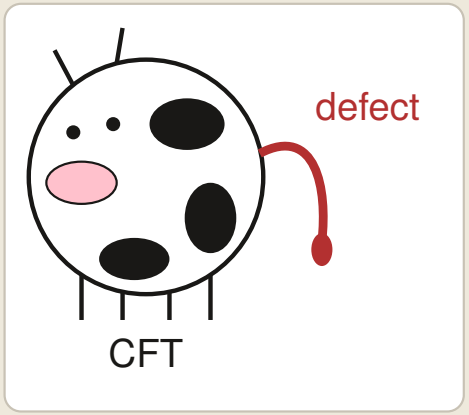
Two questions: which data, and which functions?

A tail for the spherical cow: at criticality, boundaries and impurities are never far away

[Cardy, '84][McAvity, Osborn, '95]

Away from criticality,
$\langle \mathcal{O}(x)\mathcal{O}(0) \rangle \sim e^{-|x|/\xi}$: a wall far away is invisible.

At criticality $\xi \to \infty$ and
$\langle \mathcal{O}(x)\mathcal{O}(0) \rangle \sim |x|^{-2\Delta}$: extended objects affect correlators at all distances.



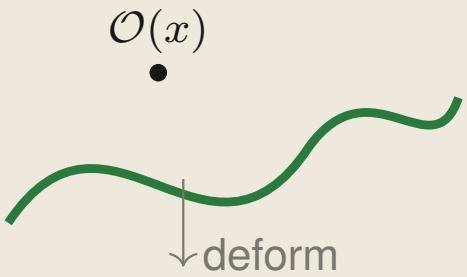
Examples of defects, on lines, boundaries and surfaces:

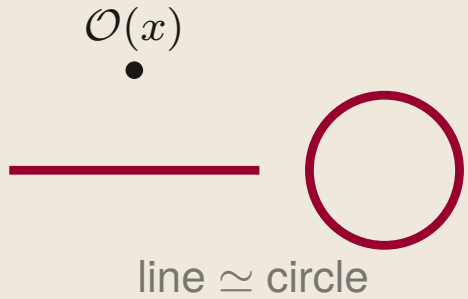
- ✕ an impurity atom in a crystal: a line along time;
- ✕ the surface of a material: a boundary;
- ✕ a heavy quark in a gauge theory: a Wilson line.

Defects are the tail of the spherical cow: details that survive at large distances.

Topological defects can be deformed freely; conformal defects only by conformal transformations

[Carqueville, Del Zotto, Runkel, '23][Billò, Gonçalves, Lauria, Meineri, '16]





Topological: correlators do not change when the support is deformed away from other insertions.

Conformal: a p-dimensional defect preserves $SO(d+1,1) \to SO(p+1,1) \times SO(d-p)$.

A topological defect is conformal, but a conformal defect is rarely topological.

A conformal defect carries its own spectrum of local operators, and a displacement operator

[McAvity, Osborn, '95][Billò, Gonçalves, Lauria, Meineri, '16]

Defect operators $\hat{\mathcal{O}}(\tau)$ have scaling dimensions $\hat{\Delta}$, and bulk operators expand on the defect:

$$\mathcal{O}_\Delta(x) = \sum_{\hat{\mathcal{O}}} \frac{b_{\mathcal{O}\hat{\mathcal{O}}}}{|x_\perp|^{\Delta-\hat{\Delta}}}\, \hat{\mathcal{O}}(x_\parallel) + \dots$$

Translations orthogonal to the defect are broken, which defines the displacement operator:

$$\partial_\mu T^{\mu i}(x) = \delta^{(d-p)}(x_\perp)\, \mathbb{D}^i(x_\parallel)\,, \qquad \hat{\Delta}_{\mathbb{D}} = p+1\,.$$

A defect is topological when its displacement operator vanishes: moving it costs nothing.

Topological and conformal defects: what they share and where they differ

[Carqueville, Del Zotto, Runkel, '23][Mulevičius–Runkel, '20][Billò, Gonçalves, Lauria, Meineri, '16]

	Topological	Conformal
Support	deformed freely	deformed only conformally
Displacement	$\mathbb{D} = 0$	$\mathbb{D} \neq 0$, $\hat{\Delta}_{\mathbb{D}} = p + 1$
Point operators on it	topological, $\hat{\Delta} = 0$, form an algebra	a spectrum $\hat{\Delta} \geq 0$: a 1d CFT on a line
Fusion	regular, categorical	singular, with a Casimir energy
Role	symmetries, dualities	probes, impurities, charged objects
Data	fusion rules, F-symbols	$\hat{\Delta}$, $b_{\mathcal{O}\hat{\mathcal{O}}}$, $\hat{\lambda}_{\hat{\mathcal{O}}_1\hat{\mathcal{O}}_2\hat{\mathcal{O}}_3}$

Shared: both are labelled by algebraic data, carry algebras of point operators, and are connected by flows (condensation, or defect RG).

The same holonomy is topological in Chern-Simons theory and conformal in $\mathcal{N} = 4$ SYM

[Wilson, '74][Witten, '89][Reshetikhin, Turaev, '91][Maldacena, '98][Marino, Pretto, Sperling, '26]

Given a gauge field A, a representation $\mathcal{R}$ of the gauge group and a loop $\mathcal{C}$:

$$W_{\mathcal{R}}[\mathcal{C}] = \mathrm{tr}_{\mathcal{R}}\ \mathbb{P}\exp\left(i \oint_{\mathcal{C}} A\right).$$

- × Yang–Mills: a heavy quark, $\langle\, W[T \times L]\,\rangle \sim e^{-T\,V(L)}$. An area law signals confinement.
- × Chern–Simons: no metric, $W_{\mathcal{R}}$ is topological; its correlators are knot invariants, and the lines form a modular tensor category.
- × $\mathcal{N} = 4$ SYM: the line is conformal, and $V(L) \propto -1/L$ at any coupling.

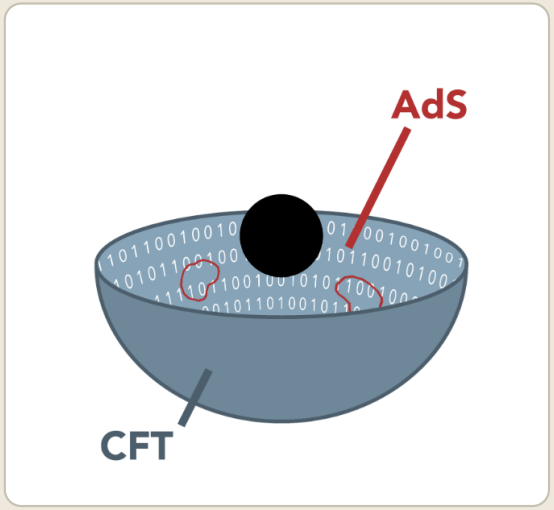
Same object, different theory:
we study the conformal case.

$\mathcal{N} = 4$ super Yang-Mills is a CFT at every coupling, and is dual to strings in $AdS_5 \times S^5$

[Maldacena, '97][Witten, '98]

- ✕ Fields in the adjoint of $\mathfrak{g}$: a gauge field A_μ, six scalars Φ^I, four Weyl fermions.
- ✕ $PSU(2,2|4)$ superconformal symmetry, $SO(6)$ R-symmetry.
- ✕ Exact tools: integrability, localisation, bootstrap.

Holography: the 4d CFT is equivalent to a gravitational theory in one more dimension. Strong coupling in the CFT is weak curvature in AdS.



drawing: J. Barrat

An interacting gauge theory where many questions can be answered exactly.

The Maldacena-Wilson line couples to one scalar and is a half-BPS conformal line defect

[Maldacena, '98][Erickson, Semenoff, Zarembo, '00][Pestun, '07]

$$W_{\mathcal{R}} = \frac{1}{\dim \mathcal{R}} \operatorname{tr}_{\mathcal{R}} \mathbb{P} \exp \int_{\mathcal{C}} ds \left(i A_\mu \dot{x}^\mu + |\dot{x}| \, \theta^I \Phi_I \right), \qquad \theta^I \theta^I = 1 \,.$$

For a straight line or a circle:

- ✕ it preserves half of the supercharges, $PSU(2,2|4) \to OSp(4^*|4)$;
- ✕ the bosonic part $\mathfrak{sl}(2) \oplus \mathfrak{su}(2) \oplus \mathfrak{sp}(4)$ is 1d conformal, transverse rotations, and $SO(5)$ R-symmetry;
- ✕ $\langle W \rangle = 1$ for the line; for the circle it is computed exactly by a matrix model.

A conformal line defect with a lot of symmetry: the starting point.

Insertions on the line form a one-dimensional CFT; at strong coupling, a string worldsheet in AdS_2

[Drukker, Kawamoto, '06][Cooke, Dekel, Drukker, '17][Giombi, Roiban, Tseytlin, '17]

Correlators of insertions $\mathcal{O}_i(t_i)$ on the straight line:

$$\langle\, \mathcal{O}_1(t_1) \cdots \mathcal{O}_n(t_n) \,\rangle = \frac{\langle\, \text{tr}\, \mathbb{P}\big[\mathcal{O}_1(t_1) \cdots \mathcal{O}_n(t_n)\, W\big] \,\rangle}{\langle\, \text{tr}\, W \,\rangle}\,.$$

- ✕ 1d conformal symmetry: dimensions, OPE coefficients, crossing; the insertions are ordered.
- ✕ Strong coupling: fluctuations of the string ending on the line.

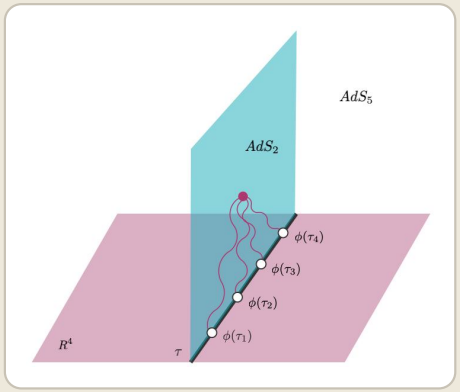


figure: G. Peveri

A 1d CFT with a lot of symmetry: the laboratory of this talk.

Operators on the line organise into $OSp(4^*|4)$ multiplets, some of them protected

[Liendo, Meneghelli, Mitev, '18][Ferrero, Meneghelli, '24]

Labels: dimension Δ, $\mathfrak{su}(2)$ spin s, $\mathfrak{sp}(4)$ Dynkin labels $[a,b]$.

- ✕ Long $L_s^{\Delta}[a,b]$, with $\Delta > a+b+2s+1$: dimensions depend on g_{YM}.
- ✕ Short $B[a,b]$, with $\Delta = a+b$: quarter-BPS, can recombine into long multiplets.
- ✕ Half-BPS $\mathcal{D}_k = B[0,k]$: absolutely protected, $\Delta = k$ at any coupling.

The displacement multiplet $\mathcal{D}_1$: the tilt $\rho(\Phi^I)$, $I = 1,\dots,5$, with $\Delta = 1$, up to the displacement, with $\Delta = 2$.

Protected multiplets can be counted in the free theory.

The question: which local operators live on a Wilson line in a general representation, and what do they do to it?

[**DA**, Meneghelli, Savi, Treilis, '26]

Which protected operators live on the half-BPS Wilson line in a representation $\mathcal{R}$, and do the new ones deform the line?

- Q1 What is the algebraic structure of the protected insertions, and how many are there?
- Q2 The exotic displacements define marginal deformations. Are they relevant or irrelevant?
- Q3 What happens when we deform the ordinary, non-supersymmetric line instead?

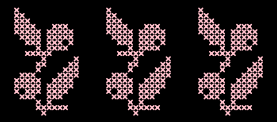
In short: a non-commutative algebra of insertions, a beta function fixed by OPE data, and a stable half-BPS line

[**DA**, Meneghelli, Savi, Treilis, '26]

- A1 Half-BPS insertions form Kirillov's family algebra $(\mathbb{C}[\mathfrak{g}] \otimes \mathrm{End}\,\mathcal{R})^G$. A Hilbert series counts them, and there are $\mathfrak{m}_{\mathcal{R}}$ copies of the displacement multiplet.
- A2 The beta function of the exotic deformations is fixed by OPE coefficients, at any g_{YM}. They are marginally irrelevant: the half-BPS line is infrared stable for every $\mathcal{R}$.
- A3 From the ordinary line, the exotic directions open new RG flows, with new fixed points.



NEW OPERATORS ON THE LINE





On the line, local operators need only be covariant: they are $\dim \mathcal{R} \times \dim \mathcal{R}$ matrices

[Drukker, Kawamoto, '06][**DA**, Meneghelli, Savi, Treilis, '26]

An open line transforms as $W_{x_i \to x_f} \to \rho_{\mathcal{R}}(g(x_i))\, W_{x_i \to x_f}\, \rho_{\mathcal{R}}(g(x_f))^{-1}$. So an insertion $\mathcal{O} \in \mathrm{End}(\mathcal{R})$ needs only

$$\boxed{\mathcal{O}[A^g, \Phi^g] = \rho_{\mathcal{R}}(g)\, \mathcal{O}[A, \Phi]\, \rho_{\mathcal{R}}(g)^{-1}}$$

instead of gauge invariance.

This condition has far more solutions than gauge invariance in the bulk.

Half-BPS insertions are equivariant polynomial maps: they form Kirillov's family algebra

[Kirillov, '00][Hausel, '24][**DA**, Meneghelli, Savi, Treilis, '26]

Half-BPS superprimaries are built from a single letter $X = u^I \Phi_I \in \mathfrak{g}_{\mathbb{C}}$, with $u \cdot u = 0$:

$$\mathcal{O}[gXg^{-1}] = \rho_{\mathcal{R}}(g)\, \mathcal{O}[X]\, \rho_{\mathcal{R}}(g)^{-1} \implies \mathcal{M}_{\mathcal{R}} = \left(\mathbb{C}[\mathfrak{g}] \otimes \mathrm{End}\, \mathcal{R}\right)^G,$$

graded by the degree k in X.

- ✕ In the bulk: $\mathbb{C}[\mathfrak{g}]^G = \mathbb{C}[P_{d_1}, \ldots, P_{d_r}]$, the Coulomb-branch chiral ring: commutative, freely generated by Casimirs.
- ✕ On the line: $\mathcal{M}_{\mathcal{R}}$ contains it as central elements $P(X)\mathbf{1}$, and it is a free module extension in general not commutative.
- ✕ Degree zero: $\mathcal{M}_{\mathcal{R},0} = \mathrm{End}_G(\mathcal{R}) = \mathbb{C}$ by Schur. The line is simple.

A Hilbert series counts the half-BPS insertions, and its linear term counts the exotic displacements

[Hanany,Sperling, '16][**DA**, Meneghelli, Savi, Treilis, '26]

dim $\mathcal{M}_{\mathcal{R},k}$ is the multiplicity of the trivial representation in $\mathcal{R} \otimes \bar{\mathcal{R}} \otimes \mathrm{Sym}^k(\mathrm{Adj})$. With the Weyl integration formula,

$$H_{\mathcal{R}}(x) = \sum_{k\geq 0} \dim \mathcal{M}_{\mathcal{R},k}\, x^k = \frac{(1-x)^{-r}}{|W|} \oint_{\mathbb{T}^r} \prod_{a=1}^{r} \frac{dz_a}{2\pi i z_a}\, \chi_{\mathcal{R}} \chi_{\bar{\mathcal{R}}} \prod_{\alpha\in\Delta} \frac{1-e^{\alpha}}{1-x\, e^{\alpha}} \,.$$

- ✕ It is rational: $H_{\mathcal{R}}(x) = P_{\mathcal{R}}(x) / \prod_a (1-x^{d_a})$, e.g. $P^{\mathfrak{su}(2)}_{[\lambda]} = \sum_{l=0}^{\lambda} x^l$.
- ✕ Universally $H_{\mathcal{R}}(x) = 1 + \mathfrak{m}_{\mathcal{R}}\, x + \ldots$: there are $\mathfrak{m}_{\mathcal{R}}$ multiplets $\mathcal{D}_1$.
- ✕ At large N, symmetric and antisymmetric lines have the same half-BPS spectrum.

Moving the line inserts the field strength, and in a rich representation there are many inequivalent ways to do so

[**DA**, Meneghelli, Savi, Treilis, '26]

A small deformation $x^\mu \to x^\mu + \delta x^\mu$ of the line inserts $\int ds\, \rho_{\mathcal{R}}(F_{\mu\nu})\, \dot{x}^\mu \delta x^\nu$. But for a general $\mathcal{R}$ there are

$$\mathcal{O}^{(i)}(F_{\mu\nu}) \in \mathrm{End}(\mathcal{R})\,, \qquad i = 0, \dots, \mathfrak{m}_{\mathcal{R}} - 1\,,$$

with $\mathfrak{m}_{\mathcal{R}}$ the number of non-zero Dynkin labels of $\mathcal{R}$. For $\mathfrak{su}(N)$:

$$\mathcal{O}^{(i)}(X) = \left(\rho_{\mathcal{R}}(T^a)\, \frac{\partial}{\partial X^a}\right)^{i+1} \mathrm{tr}\left(X^{i+2}\right).$$

These are the exotic operators, with the same quantum numbers as the standard $\mathcal{O}^{(0)} \propto \rho_{\mathcal{R}}$.

The exotic displacements commute, but higher products depend on the order: a one-dimensional effect

[Okubo, Myung, '81][Hausel, '24][**DA**, Meneghelli, Savi, Treilis, '26]

Degree one is spanned by $\mathcal{O}_1^{(i)}(X) = \mathcal{O}_{1a}^{(i)} X^a$, $i = 0, \dots, \mathfrak{m}_{\mathcal{R}} - 1$. With the same polarisation, the order does not matter:

$$\left[\mathcal{O}_1^{(i)}(X), \mathcal{O}_1^{(j)}(X)\right] = 0\,.$$

With different polarisations, the $[X, Y]$ channel of their OPE defines

$$i f^{ab}{}_c \, \mathcal{O}_{1a}^{(i)} \mathcal{O}_{1b}^{(j)} = \mu_{ij}{}^k \, \mathcal{O}_{1c}^{(k)}\,,$$

a commutative, associative product.

A Frobenius algebra with unit the standard displacement.

Quarter-BPS insertions can recombine, but $\mathfrak{m}_{\mathcal{R}}(\mathfrak{m}_{\mathcal{R}}-1)$ of them stay protected

[Ferrero, Meneghelli, '24][**DA**, Meneghelli, Savi, Treilis, '26]

In the free theory, two letters $X, Y \in \mathfrak{g}$ give $\mathfrak{m}_{\mathcal{R}}^2$ multiplets $B_1[2,0]$:

$$\mathcal{O}^{(i,j)}_{(2,0)}(X,Y) = \mathcal{O}^{(i)}_1(X)\,\mathcal{O}^{(j)}_1(Y) - \mathcal{O}^{(i)}_1(Y)\,\mathcal{O}^{(j)}_1(X)\,.$$

For $g_{\text{YM}} > 0$, the $\mathfrak{m}_{\mathcal{R}}$ combinations $\mathcal{O}^{(0,i)} + \mathcal{O}^{(i,0)} = 2\,\mathcal{O}^{(i)}_1([X,Y])$ are $\mathfrak{Q}$-exact and recombine. The other $\mathfrak{m}_{\mathcal{R}}(\mathfrak{m}_{\mathcal{R}}-1)$ remain protected, half even and half odd under 1d parity: $B_1[2,0]^{\pm}$.

These protected operators will control the beta function.

A conjecture: protected quarter-BPS insertions are equivariant maps modulo commutators

[Cachazo, Douglas, Seiberg, Witten, '02][Grant, Grassi, '08][**DA**, Meneghelli, Savi, Treilis, in prep.]

In the free theory, quarter-BPS superprimaries are built from two letters $X, Y \in \mathfrak{g}_{\mathbb{C}}$:

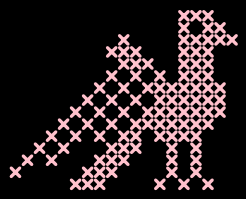
$$\mathcal{M}^{\text{free}}_{\frac{1}{4}} = \left(\mathbb{C}[\mathfrak{g} \oplus \mathfrak{g}] \otimes \text{End}\,\mathcal{R}\right)^{G},$$

graded by $\mathbb{C}^* \times SU(2)$: the total degree, and the doublet (X, Y).

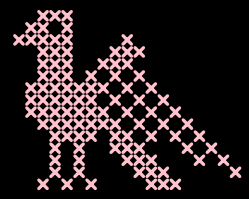
For $g_{\text{YM}} > 0$ commutators are $\mathfrak{Q}$-exact, as $\mathcal{O}_1^{(i)}([X, Y])$ on the previous slide. The conjecture:

$$\boxed{\mathcal{M}_{\frac{1}{4}} \simeq \left(\mathbb{C}[\mathfrak{g} \oplus \mathfrak{g}]/\langle [X, Y] \rangle \otimes \text{End}\,\mathcal{R}\right)^{G}}$$

- ✕ $\mathcal{R}$ trivial: the Higgs-branch chiral ring $\left(\mathbb{C}[\mathfrak{g} \oplus \mathfrak{g}]/\langle [X, Y] \rangle\right)^{G}$. In general, a module over it; $Y = 0$ gives back $\mathcal{M}_{\mathcal{R}}$.
- ✕ Degree two, $SU(2)$ singlet: $\mathfrak{m}_{\mathcal{R}}^2 - \mathfrak{m}_{\mathcal{R}} = \mathfrak{m}_{\mathcal{R}}(\mathfrak{m}_{\mathcal{R}} - 1)$ multiplets $B_1[2, 0]$, none for the fundamental line.



DEFORMING THE LINE





Each exotic displacement defines a marginal deformation of the half-BPS line

[**DA**, Meneghelli, Savi, Treilis, '26]

$$W \;\to\; W \,\exp\left(\zeta^{\mathcal{I}} \int dt\, D_{\mathcal{I}}(t)\right), \qquad D_{(i,I)} = \frac{2\pi}{g_{\text{YM}}}\, \mathcal{O}^{(i)}_{1a}\, \Phi^a_I + \dots$$

with $i = 0, \dots, \mathfrak{m}_{\mathcal{R}} - 1$ and $I = 1, \dots, 5$: $5\mathfrak{m}_{\mathcal{R}}$ couplings.

- ✗ $\Delta = 1$ at $\zeta = 0$, for every g_{YM}: the deformations are marginal.
- ✗ $i = 0$ rotates θ^I: it moves along the family of half-BPS lines.
- ✗ $i \geq 1$ break all supersymmetry: the exotic directions.

Marginal is not exactly marginal: do the exotic couplings grow or decay?

In conformal perturbation theory the beta function is an integrated four-point function

[Komargodski, Simmons-Duffin, '16][Behan, Rastelli, Rychkov, Zan, '17]

Three-point functions of $D_{\mathcal{I}}$ vanish by $\mathfrak{sp}(4)$, so the first correction is cubic:

$$\beta^{\mathcal{I}} = \frac{d\zeta^{\mathcal{I}}}{d\log(1/a)} = -\frac{1}{3}\,\zeta^{\mathcal{J}}\zeta^{\mathcal{K}}\zeta^{\mathcal{L}}\,\delta_{\mathcal{J}}C_{\mathcal{K}\mathcal{L}}{}^{\mathcal{I}} + \dots$$

with a a short-distance cutoff and

$$\frac{\delta_{\mathcal{I}}C_{\mathcal{J}\mathcal{K}\mathcal{L}}}{t_{32}\,t_{31}\,t_{21}} = \int dt\,\langle\, D_{\mathcal{I}}(t)\,D_{\mathcal{J}}(t_1)\,D_{\mathcal{K}}(t_2)\,D_{\mathcal{L}}(t_3)\,\rangle\,.$$

We need an integral of a four-point function at finite coupling.

Superconformal symmetry turns the integral into boundary terms, that is, into OPE coefficients

[Liendo, Meneghelli, Mitev, '18][**DA**, Meneghelli, Savi, Treilis, '26]

Superconformal Ward identities fix the four-point function up to a constant and one function:

$$\langle D^{(i)} D^{(j)} D^{(k)} D^{(\ell)} \rangle = (13)(24)\left(\mathbb{F}_{ijk\ell} - \partial_\chi\left(\frac{(\chi-y_1)(\chi-y_2)}{\chi^2} f_{ijk\ell}(\chi)\right)\right).$$

The integrand is a total derivative: only the OPE limits contribute. Identity and $\mathcal{D}_2$ cancel, and

$$\delta_{\mathcal{I}} C_{\mathcal{J}\mathcal{K}\mathcal{L}} = -2\,\delta_{IJ}\delta_{KL}\left(C^2_{B_1[2,0]^+}\right)_{ijk\ell} + 2 \text{ permutations},$$

$$\left(C^2_X\right)_{ijk\ell} = \sum_{\mathcal{O},\mathcal{O}'\in X} C_{ij}{}^{\mathcal{O}}\, C_{k\ell}{}^{\mathcal{O}'}\, g_{\mathcal{O}\mathcal{O}'}\,.$$

A selection rule fixes the beta function in terms of quarter-BPS OPE data, at any g_{YM}

[**DA**, Meneghelli, Savi, Treilis, '26]

Choosing one index along the super-displacement, $\mathfrak{Q}$-cohomology gives

$$C_{0i}{}^{\mathcal{O}} = 0 \qquad \forall\, \mathcal{O} \in B_1[2,0]^+ \,.$$

Then

$$\boxed{\beta^{(i,I)} = 2\,\zeta^{(\ell,I)}\,(\zeta^j \cdot \zeta^k)\,\big(C^2_{B_1[2,0]^+}\big)_{jk,\ell i'}\,g^{i'i} + O(\zeta^5)}$$

with $\cdot$ the $SO(5)$ product. At this order it is a gradient flow.

Non-perturbative in g_{YM}: only protected OPE data enter.

The exotic deformations are marginally irrelevant: the half-BPS line is infrared stable for every representation

[**DA**, Meneghelli, Savi, Treilis, '26]

Distance from the half-BPS locus: $R^2_{\text{ex}} = g_{\hat{i}\hat{j}}\,\zeta^{\hat{i}} \cdot \zeta^{\hat{j}}$, over the exotic directions only. In an orthonormal basis of $B_1[2,0]^+$,

$$\frac{1}{2}\frac{dR^2_{\text{ex}}}{d\log(1/a)} = 2\sum_A \Big[\sum_{\hat{i},\hat{j}} (\zeta^{\hat{i}} \cdot \zeta^{\hat{j}})\, C^A_{\hat{i}\hat{j}}\Big]^2 + O(\zeta^6) \;\geq\; 0\,.$$

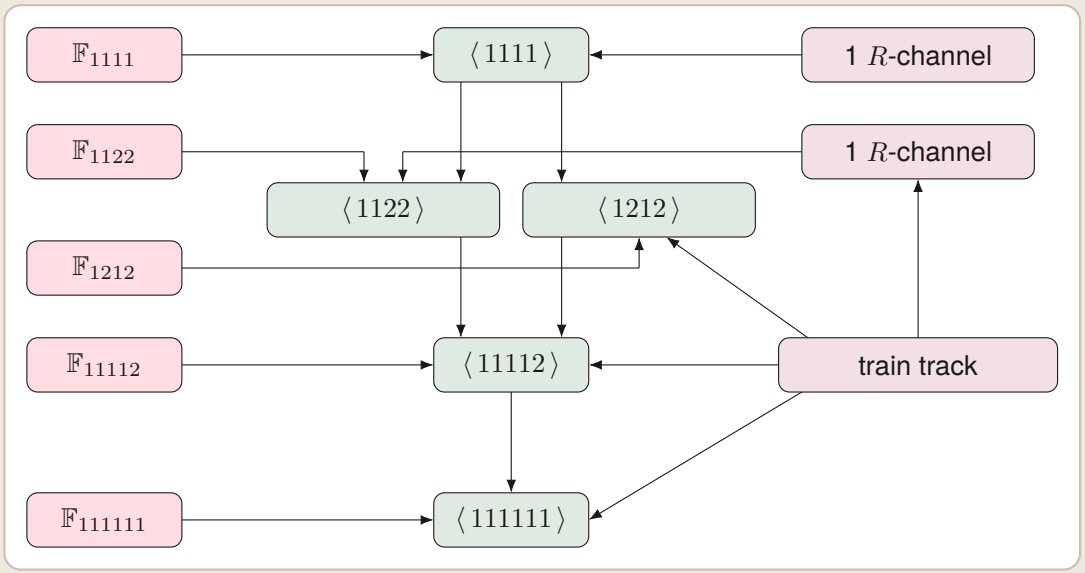
- R_{ex} shrinks towards the infrared: small exotic deformations flow back.
- Strictly positive, since $C^A_{\hat{i}\hat{j}}$ is non-degenerate for the half-BPS line.
- The argument holds for any $OSp(4^*|4)$-invariant line in $\mathcal{N}=4$ SYM.

The half-BPS Wilson line is rigid against all exotic deformations.

The perturbative bootstrap: impose all symmetry first, then compute the fewest Feynman diagrams

[**DA**, Barrat, Peveri, '24][**DA**, Barrat, Xu, '24]

1. Ward identities and crossing: few functions.
2. An ansatz of polylogarithms of uniform weight.
3. The simplest R-symmetry channel from Feynman diagrams.
4. Pinching to lower points fixes the rest.



left: localisation; right: diagrams

Each correlator is built on the lower-point ones.

Superconformal Ward identities give a topological sector and leave a single function

[Liendo, Meneghelli, Mitev, '18][Giombi, Komatsu, '18][**DA**, Barrat, Peveri, '24]

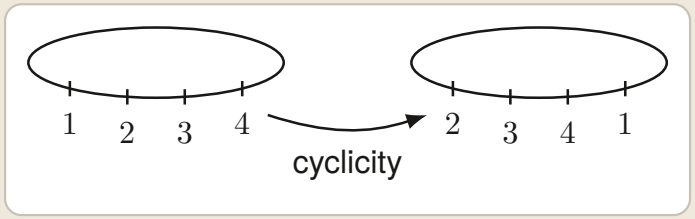
$$\left(\tfrac{1}{2}\partial_x + \alpha\,\partial_r - (1-\alpha)\,\partial_s\right)\mathcal{A}(x;r,s)\Big|_{r=\alpha x,\ s=(1-\alpha)(1-x)} = 0\,.$$

- ✕ Topological sector: $\mathcal{A}\big(x;x^2,(1-x)^2\big)=\mathbb{F}$, a number known exactly from localisation.

The solution depends on a single function $f(x)$:

$$\mathcal{A} = \frac{1}{2}\left(\frac{r}{x^2}+\frac{s}{(1-x)^2}\right)\mathbb{F} + \partial_x\left(\left(1-\frac{r}{x}-\frac{s}{1-x}\right)f(x)\right).$$

Cyclicity of the trace gives crossing, $f(x) = -f(1-x)$.
Pinching two insertions gives lower-point functions.



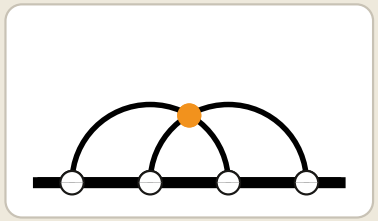
Four points to two loops: the simplest channel is non-planar at tree level and needs one diagram at one loop

[Kiryu, Komatsu, '18][Cavaglià, Gromov, Julius, Preti, '22][**DA**, Barrat, Peveri, '24]

For $\langle 1111 \rangle$, with λ the 't Hooft coupling:

$$F_1^{(0)} = 0\,,$$

$$f^{(1)}(x) = \frac{1}{8\pi^2}\left(\frac{\pi^2}{6} + G(1,0;x) - G(0,1;x)\right).$$



X diagram

At two loops: $f^{(2)}$ is a combination of weight-four polylogarithms and ζ-values with rational coefficients. Four terms of the small-x expansion of $F_1^{(2)}$ fix it.

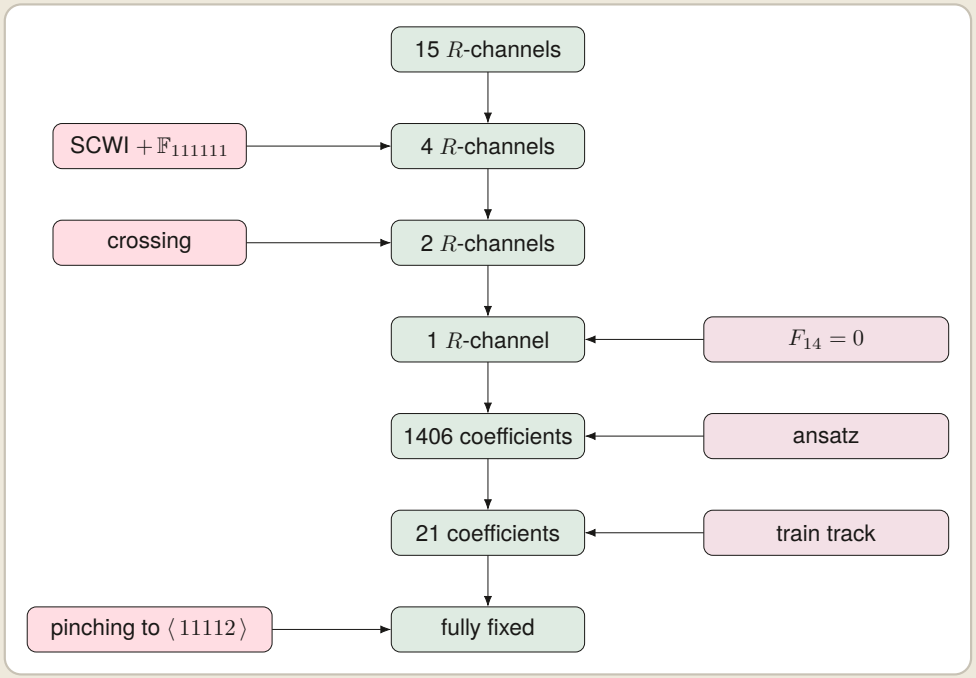
Uniform transcendentality turns a hard calculation into linear algebra.

Five and six points at two loops: symmetry leaves one function, and the train track fixes it

[**DA**, Barrat, Peveri, '24]

$\langle 11112 \rangle$: 6 channels $\to$ 1 function. Ansatz of 231 coefficients; one integral, the train track, leaves 20; pinching to $\langle 1122 \rangle$, $\langle 1212 \rangle$ fixes them.

$\langle 111111 \rangle$: 15 channels $\to$ 1 function. Ansatz of 1406 coefficients; the train track leaves 21; pinching to $\langle 11112 \rangle$ fixes them, with five more relations as checks.



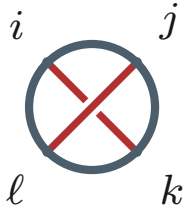
How far the method goes: six points at two loops.

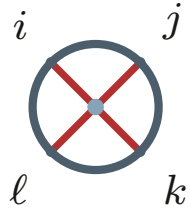
For exotic insertions the same bootstrap works, with colour tensors of the representation

[**DA**, Barrat, Peveri, '24][**DA**, Meneghelli, Savi, Treilis, '26]

The same steps for $\langle D^{(i)} D^{(j)} D^{(k)} D^{(\ell)} \rangle$, with colour structures from traces over $\mathcal{R}$, for any $\mathfrak{g}$ and $\mathcal{R}$:

$\mathsf{g}_{ij} =$ 

$\mathsf{b}_{ijk\ell} =$ 

$\mathsf{x}_{ijk\ell} =$ 

Leading order:

$$\mathbb{F}^{(0)} = \mathsf{g}_{ij}\mathsf{g}_{k\ell} + \mathsf{g}_{i\ell}\mathsf{g}_{jk} + \mathsf{b}_{ijk\ell}\,, \qquad f^{(0)} = \tfrac{\chi}{1-\chi}\big((1-\chi)\mathsf{g}_{ij}\mathsf{g}_{k\ell} - \chi\,\mathsf{g}_{i\ell}\mathsf{g}_{jk}\big)\,.$$

One more colour tensor, x, carries all the interaction at one loop.

The next-to-leading order is remarkably simple, and confirms the non-perturbative beta function

[**DA**, Meneghelli, Savi, Treilis, '26]

With $\hat{\lambda}_{\mathfrak{g}} = g^2_{\text{YM}} h^\vee_{\mathfrak{g}} / 4\pi^2$:

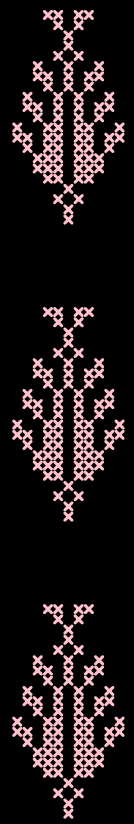
$$\mathbb{F}^{(1)}_{ijk\ell} = \frac{\pi^2}{6}\, \mathsf{x}_{ijk\ell}\,, \qquad f^{(1)}_{ijk\ell} = \frac{\chi}{1-\chi}\big(u(\chi) - u(1-\chi)\big)\, \mathsf{x}_{ijk\ell}\,,$$

with $u(\chi) = \frac{1}{2}\text{Li}_2(\chi) - \frac{\pi^2}{12}\chi$. Operators at the unitarity bound all acquire $\Delta = 1 + \hat{\lambda}_{\mathfrak{g}} + \dots$, and the block expansion gives

$$2\big(C^2_{B_1[2,0]_+}\big)^{(0)}_{ij,k\ell} = \mathsf{g}_{ij}\mathsf{g}_{k\ell} + \mathsf{g}_{ik}\mathsf{g}_{j\ell} + \mathsf{g}_{i\ell}\mathsf{g}_{jk} - \mathbb{F}^{(0)}_{ijk\ell} - \mathsf{x}_{ijk\ell}\,,$$

totally symmetric in its indices.

Total symmetry gives an even stronger stability property at weak coupling.



BEYOND THE FUNDAMENTAL REPRESENTATION

From the ordinary line, the exotic directions open new RG flows with new fixed points

[Polchinski, Sully, '11][**DA**, Meneghelli, Savi, Treilis, '26]

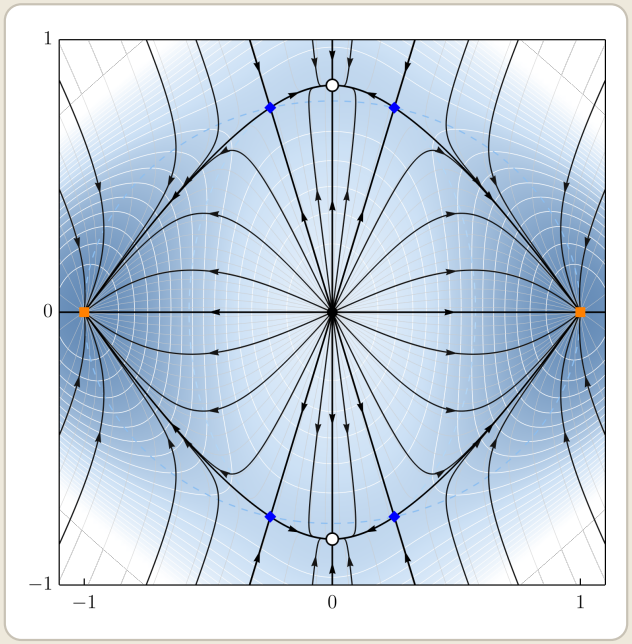
$W_\xi = \mathbb{P}\exp\int dt\,\left(iA_t + \xi^{(i,M)}\,\mathcal{O}^{(i)}_{1a}\Phi^a_M\right)$, with $M = 1,\dots,6$, has

$$\beta^{(i,M)} = -\frac{\hat{\lambda}_{\mathfrak{g}}}{2}\left[\xi^{(i,M)} + \frac{2}{h^\vee}\,\xi^{(\ell,M)}(\xi^j\cdot\xi^k)\,\mathsf{S}_{jk\ell i'}\,g^{i'i}\right].$$

For $i = 0$ only: the known flow from the ordinary ($\xi = 0$) to the half-BPS line ($|\xi| = 1$).

Adjoint of $SU(3)$, $\mathfrak{m}_{\mathcal{R}} = 2$, in a plane:

- ✕ half-BPS lines: IR stable;
- ✕ exotic lines $(0, \pm 3/\sqrt{13})$: stable in the plane, unstable outside;
- ✕ mixed points $(\pm\frac{1}{4}, \pm\frac{3}{4})$: one relevant direction;
- ✕ a generic deformation still flows to the half-BPS line.



Summary: the spectrum of a Wilson line is much richer than in the bulk, and the half-BPS line is rigid

[**DA**, Meneghelli, Savi, Treilis, '26]

Q1 *Which protected insertions?*
Half-BPS ones form the family algebra $(\mathbb{C}[\mathfrak{g}] \otimes \mathrm{End}\,\mathcal{R})^G$, graded and non-commutative; a Hilbert series counts them; $\mathfrak{m}_{\mathcal{R}}$ copies of the displacement multiplet.

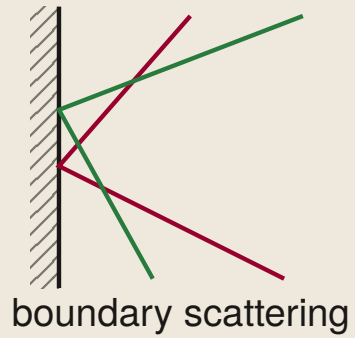
Q2 *Relevant or irrelevant?*
Marginally irrelevant. The beta function is fixed by quarter-BPS OPE data at any coupling, and the half-BPS line is infrared stable for every $\mathcal{R}$.

Q3 *And from the ordinary line?*
New fixed points appear, but generic flows still end on the half-BPS line.

Outlook: is a conformal fixed point of a Wilson line also integrable?

[Drukker, Kawamoto, '06][Correa, Leoni, Luque, '18][Correa, Schaposnik Massolo, '13]

On the fundamental line, the one-loop mixing of insertions is an open spin chain, solvable when the boundary Yang–Baxter equation holds. The deformed fundamental line is one-loop integrable only at its conformal fixed points.



For an exotic sector, e.g. in the representation $(1, 1, 0, \ldots, 0)$ of $SU(N)$:

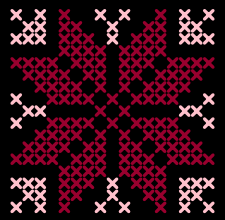
- ✗ one-loop mixing: an open chain with exotic boundary degrees of freedom;
- ✗ fixed points from the ζ^5 term of the beta function, i.e. integrated six-point functions, and the boundary Yang–Baxter test at each.

Either an integrable exotic sector, or the interaction that breaks integrability.

Other directions: higher orders, strong coupling, other defects

[Gomis, Passerini, '06][Meneghelli, Trépanier, '23]

- ✕ Higher orders: the ζ^5 term of the beta function, from integrated six-point functions of exotic insertions.
- ✕ Holography: lines in large representations are D3/D5 branes in $AdS_5 \times S^5$; the exotic multiplets should be their moduli.
- ✕ Exact methods: localisation in the topological sector, and a superconformal index for the line.
- ✕ Algebra: test the quarter-BPS conjecture; closed Hilbert series beyond $\mathfrak{su}(2)$, $\mathfrak{su}(3)$; combinatorics of $P_{\mathcal{R}}(x)$.
- ✕ Other defects: half-BPS surface defects of the 6d $(2,0)$ theory have a similar multitude of operators.





THANK YOU!

